

Delphi-Psychometric Evaluation of 3D Generative Models: Combining Structured LLM Elicitation with Item Response Theory

Suki Wang

Roblox

siqiwang@roblox.com

Stanford AIMS Workshop on AI Measurement Science @ COLM 2026

Abstract. Benchmarks for 3D generative models aggregate automated metrics without principled analysis of reliability, item difficulty, or judge consistency. We introduce a framework combining three previously disconnected methodologies: **Delphi-structured multi-round LLM judging** (for scalable, calibrated elicitation), **Item Response Theory (IRT)** and **Many-Facet Rasch Modelling (MFRM)** (for psychometric analysis of the resulting data), applied to Hi3DBench, a public benchmark spanning six image-to-3D generation models and 510 prompts across five quality criteria. We report five empirical contributions: (E1) the five object-level criteria load onto a single latent quality factor (1-factor CFA: CFI=0.95, RMSEA=0.09), validating within-level aggregation; a cross-level analysis adding Hi3DEval’s four material-scoring dimensions confirms they form a psychometrically distinct factor (2F CFI=0.883 vs. 1F CFI=0.641, $\Delta\text{BIC} = -1,763$), so object-level and material-level scores should not be combined; (E2) evaluation criterion, treated as the fixed benchmark-specified factor it is, accounts for 82% of score variance (η^2 ; a random-effects estimate agrees at 84.7%)—dwarfing model (1.4%) and prompt (0.9%), with previously-omitted model $\times$ criterion and prompt $\times$ criterion interactions comparably small (2.1%, 1.5%); per-criterion standardization directly confirms this is a scale artefact of the automated metrics, collapsing criterion’s share to $\approx 0\%$ and lifting model’s share nearly 10-fold. Delphi LLM judging on a uniform 0–9 scale cross-validates this finding by making model quality the dominant source (37%), and achieves strong reliability (ICC= 0.81–0.88; inter-judge SD reduced 68% across Delphi rounds); (E3) in a flipped 2PL IRT orientation (prompts as persons, models as items, $N = 470$ eligible prompts), model stringency spans a > 8 -logit range (HunyuanDiT-3D $b = -5.29$ to CRM $b = 2.83$); prompt generatability $\hat{\theta}_p \in [-1.07, 1.14]$ with 75 near-ceiling prompts ($p \geq 0.8$); (E4) descriptive sensitivity analysis reveals 71.6% of items show $|\text{gap}| \geq 0.40$ in pass rates between TripoSR and the five Objaverse-trained models; because Group B contains exactly one model, Objaverse exposure and TripoSR-specific factors are perfectly confounded and cannot be separated; (E5) an IRT- θ -stratified 30-item *tinyEval3D* subset recovers the full-benchmark model ranking exactly (Spearman $\rho = 1.0$, LOO-CV $\rho = 0.954$), reducing evaluation cost by 94%, with model rankings confirmed by Delphi LLM judging ($\rho = 0.90$ agreement with automated metric rankings). Our framework exposes structural inefficiencies in current 3D benchmarks and provides a reusable methodology for constructing psychometrically grounded evaluations.

1. Introduction

The rapid proliferation of text-to-3D and image-to-3D generative models has produced an equally rapid proliferation of evaluation benchmarks [Duggal et al., 2025, Zhang et al., 2025, He et al., 2023, Maiti et al., 2025]. These benchmarks share a common methodology: aggregate automated metric scores into weighted sums, report raw percentage-agreement with human judgment, and treat all evaluation items as equally informative. This methodology has well-known limitations in adjacent fields. Sixty years of psychometric research in educational testing have established that: (i) score aggregation is only valid when items measure a unidimensional construct, (ii) items differ systematically in difficulty and discriminability, (iii) rater effects must be separated from item effects, and (iv) a small set of maximally informative items can often substitute for large benchmarks. **Item Response Theory (IRT)** formalizes (ii) by modeling the probability of a correct/passing response as a function of a latent person ability and item-specific difficulty and discrimination parameters, rather than treating every item as an equally-weighted point [?]. **Many-Facet Rasch Measurement (MFRM)** extends this to designs with additional measurement facets beyond persons and items—most relevantly, raters—so that a rater’s severity is estimated and separated from the quantity being measured rather than folded into it [??]. We apply both: IRT (via a graded-response and a 2PL model) to model quality items, and an MFRM-style crossed random/fixed-effects decomposition to separate model, prompt, criterion, and judge facets.

Parallel work in text-LLM evaluation has begun applying psychometric methods [Polo et al., 2024]. However, this work has not reached 3D generative evaluation, where the multi-modal, multi-view nature of the task and the absence of

ground-truth labels make reliability analysis especially critical.

At the same time, LLM-as-judge approaches have become standard in generative AI evaluation [Zheng et al., 2023], but are typically applied as one-shot prompts without any structured disagreement resolution. When the same judge is queried repeatedly, or when judges are shown each other’s outputs with identity visible, they are susceptible to *authority bias*—conforming to a higher-status model’s assessment—and *anchor conformity*—converging toward a revealed group median without genuinely re-examining the stimulus. The Delphi method [Dalkey and Helmer, 1963], developed at RAND in the 1960s for policy forecasting, provides a protocol for iterative expert consensus that directly addresses both failure modes and has never been applied to LLM evaluation.

This paper contributes the **Delphi-Psychometric triangle**: Delphi structures the elicitation; LLMs scale it; psychometrics analyses the resulting data. We demonstrate the framework on Hi3DBench [Zhang et al., 2025], producing five empirical findings (E1–E5) that reveal structural properties of the benchmark invisible to standard analysis.

2. Related Work

3D generative evaluation. Eval3D [Duggal et al., 2025] introduces five automated metrics covering geometric, structural, semantic, aesthetic, and text-alignment dimensions. Hi3DEval [Zhang et al., 2025] extends this with hierarchical human-aligned scoring across object- and part-level properties. T3Bench [He et al., 2023] and Gen3DEval [Maiti et al., 2025] provide complementary text-to-3D benchmarks. All use raw metric averages; none apply psychometric analysis.

Psychometrics for AI evaluation. tinyBenchmarks [Polo et al., 2024] applies Item Response Theory to text-LLM benchmarks, showing that ~ 100 items can recover full-benchmark rankings. None of these works addresses 3D evaluation.

LLM-as-judge and Delphi. LLM-as-judge / MT-Bench [Zheng et al., 2023] demonstrates LLM raters as scalable alternatives to human evaluation. The Delphi method [Dalkey and Helmer, 1963, Rowe and Wright, 1999] is an established protocol for iterative expert consensus in high-uncertainty domains. To our knowledge, Delphi has not previously been applied to LLM judging.

3. Methods

3.1. Dataset: Hi3DBench

We use Hi3DBench [Zhang et al., 2025], a public benchmark comprising rendered images of 3D assets generated by six models: CRM, HunyuanDiT-3D, SPARD, TRELLIS, TripoSR, and Unique3D. We use 510 prompts from the shared prompt library, yielding $510 \times 6 = 3,060$ (prompt, model) pairs. Each pair is evaluated on five quality criteria, aligned to Eval3D dimensions: **geometric consistency**, **structural consistency**, **semantic consistency**, **aesthetics**, and **text-3D alignment**. Psychometric analyses (E1–E5) use the Hi3DBench automated scores (geometry_score, geo_detail_score, geo_texture_score, texture_score, alignment_score) mapped to Eval3D criteria; these were validated against human judgments in the original paper (Table 1, Zhang et al. 2025). For Delphi LLM calibration, we sample 100 prompts from the same library and apply the protocol to the five models with complete rendered archives (HunyuanDiT-3D, SPARD, TRELLIS, TripoSR, Unique3D), yielding 500 (prompt, model) pairs rated by three LLM judges across up to three rounds.

3.2. Delphi-Structured LLM Judging Protocol

To collect LLM-based ratings, we implement a three-round Delphi protocol with three vision-capable LLM judges: **Claude Opus 4.6** (Anthropic), **GPT-4.1** (OpenAI), and **Gemini 2.5 Pro** (Google).

Round structure. In Round 1 (independent rating), each judge receives a multi-view rendering of the asset along with the text prompt, criteria definitions, and VQA questions for the text-3D alignment dimension, and returns a structured JSON rating (0–9 integer scale per criterion) with chain-of-thought reasoning.

In Round 2 (informed re-rating), each judge sees its own Round 1 scores, the group median and IQR across judges, an anonymized dissent summary generated by a fourth Claude Opus 4.7 call, and the Hi3DBench automated score

for the same asset (framed as a baseline to consider, not a target to match; judges are asked to explain rather than revise when their rating diverges by more than 2 points). Judges may revise or confirm each score; agreement with the automated score is never required or scored.

Round 3 follows the same structure as Round 2 with Round 2 statistics. We apply the [von der Gracht \[2012\]](#) stopping rule: if inter-judge SD drops below 1.0 on ≥ 4 of 5 criteria after Round 2, Round 3 is skipped to reduce cost.

Dissent summary. Between rounds, we summarize inter-judge reasoning disagreements using a dedicated LLM call (Claude Opus 4.7) with an anonymizing prompt, producing a 3–4 sentence summary that preserves disagreement content while removing judge identity.

Anti-sycophancy design. Three design choices specifically address sycophancy failure modes present in standard LLM-as-judge. First, **independent Round 1**: each judge scores the asset before seeing any other judge’s output, anchoring their position to the stimulus rather than to a social prior. Second, **distinct evaluative personas**: Claude acts as a conservative technical artist focused on surface quality and geometry, GPT as an analytical CV researcher focused on multi-view consistency, and Gemini as a UX designer focused on semantic clarity and user perception. These personas create structurally different failure modes, so genuine disagreements reflect substantive differences in evaluative framework rather than random noise. Third, **anonymized dissent**: judge identity is stripped from the inter-round summary, preventing judges from simply deferring to the model they perceive as most authoritative. Residual anchoring to the group median—visible in `group_stats_json`—cannot be eliminated by design, but requiring written justification for any revision creates an epistemic check on pure conformity. Exposing the automated score from Round 2 onward is a second, distinct anchoring channel: because every judge sees the same external reference point, part of the inter-judge convergence we report (§4.2) may reflect shared regression toward that anchor rather than purely independent deliberation, and the agreement we later report between Delphi and automated rankings is accordingly not fully independent confirmation. We retain this design because removing the anchor would require re-running the full protocol, and because the Delphi stream still disagrees with the automated stream on the top-ranked model (§5) despite the anchor pushing toward agreement—if anything, that disagreement is stronger evidence of a genuine measurement difference than it would be under a fully blind protocol.

3.3. Psychometric Analysis Pipeline

All analyses use R 4.5 with packages `mirt`, `lme4`, `lavaan`, `psych`, `difR`, and `irr`. Code and data are available at [anonymized for review].

E1 — Factor structure (Multidimensional IRT). We apply Horn’s parallel analysis, exploratory factor analysis (EFA) with oblimin rotation, confirmatory factor analysis (CFA), and graded-response MIRT to the (model $\times$ prompt) $\times$ 5-criterion score matrix, assessing whether the five criteria measure a single latent quality dimension.

E2 — Variance decomposition (MFRM). We fit stream-specific mixed models. Criterion is a fixed, benchmark-specified set of five rubric dimensions rather than a sample from a population of criteria, so we treat it as a *fixed* facet (β_c); model, prompt, judge, and round remain random facets, consistent with generalizability-theory practice for designs with a mix of fixed and random facets [e.g., ??]. Interactions of the fixed criterion factor with a random facet are themselves random effects. For the **automated-score stream** (no judges or rounds):

$$y_{mpc} = \mu + \beta_c + u_m + u_p + u_{mp} + u_{mc} + u_{pc} + \varepsilon$$

For the **Delphi stream** (three judges, up to three rounds), two additional facets enter:

$$y_{mpjcr} = \mu + \beta_c + u_m + u_p + u_j + u_r + u_{mp} + u_{mc} + u_{pc} + \varepsilon$$

In both equations m =model, p =prompt, c =criterion, u_{mp} is a model $\times$ prompt interaction, and u_{mc} , u_{pc} are model $\times$ criterion and prompt $\times$ criterion interactions (previously omitted); j =judge and r =round apply to the Delphi stream only. The fixed criterion effect’s share of total variance is reported as η^2 from a one-way ANOVA on criterion (unambiguous here since criterion is the only fixed factor); remaining facets are REML variance components from the mixed model, rescaled to sum to $1 - \eta^2$ so the full decomposition adds to 100%. As a robustness check, we also report the alternative

in which criterion is instead treated as random (matching a common but, for a fixed rubric, less appropriate convention), and a version in which each criterion’s scores are z -scored (mean 0, SD 1) before pooling, to isolate the contribution of raw scale differences across criteria from genuine model or prompt signal. We compute variance components as percentages of total variance, estimate the G-coefficient (generalizability), and quantify the Delphi convergence effect as the proportional reduction in judge-level variance from Round 1 to Round 3. Inter-rater reliability is reported as ICC(A, k)—two-way random-effects, absolute agreement, average-of- k -raters, in the McGraw–Wong [?] naming (equivalently ICC(2, k) in Shrout–Fleiss [?] notation)—the appropriate form given all three judges rate every item (a fully crossed design) and we report the reliability of the mean of three judges’ ratings, not of a single judge.

E3 — Model stringency and prompt generatability (flipped 2PL IRT). We dichotomize **geometric consistency** at the midpoint (threshold=5, sensitivity at 3,4,6,7) to produce a binary (model $\times$ prompt) matrix; this criterion alone spans a wide enough native range to support a midpoint split, unlike the other four criteria, whose automated-scoring ranges are capped well below the midpoint (§4.2) and would make any fixed threshold degenerate. CRM’s automated scores are unavailable for 263 of the 510 prompts (generation failures); these are scored as automatic fails on this criterion rather than excluded, so that a model’s inability to generate a scoreable asset for a prompt counts against it as a real benchmark outcome instead of silently shrinking its eligible sample. Because IRT stability requires $N \geq 200$ persons and we have only 6 models, we *transpose* the matrix—placing 470 eligible prompts as persons and 6 models as items—and fit a 2PL IRT model. This orientation yields stable model parameters (b_m : stringency; a_m : discrimination power) and prompt ability estimates $\hat{\theta}_p$ (generatability), computed via EAP scoring.

E4 — Training-Data Sensitivity Analysis. We group models by Objaverse training exposure (Group A: five Objaverse-trained models; Group B: one non-Objaverse model, TripoSR). Because standard inferential DIF tests (Mantel–Haenszel, logistic regression Magis et al. 2010) require $n \geq 10$ per group, we conduct a *descriptive sensitivity analysis*: for each item, we compute the gap in pass rates between the two groups ($\Delta p = p_{\text{Objaverse}} - p_{\text{non-Objaverse}}$) and report the distribution of $|\Delta p|$ as a measure of training-data sensitivity.

E5 — Adaptive 30-item subset (tinyEval3D). We compare four 30-item selection strategies: (i) IRT- θ -stratified sampling (equal items from each quintile of the estimated prompt generatability $\hat{\theta}_p$ from the flipped 2PL), (ii) random sampling (1,000 draws), (iii) classical difficulty-stratified sampling (quintiles of p -value), and (iv) top- N by point-biserial discrimination. We report Spearman ρ between subset-derived and full-benchmark model rankings, validated by leave-one-model-out cross-validation.

4. Results

4.1. E1: Factor Structure

CFA fit statistics (Table 1) show that a 1-factor model achieves CFI=0.95, RMSEA=0.087 (SRMR=0.037), consistent with a primarily unidimensional latent quality construct; the 1-factor model also wins on CFA BIC ($\Delta\text{BIC} = +6.7$ for 2F vs. 1F). EFA and graded-response MIRT both favour a 2-factor solution (EFA BIC= -3.87 for 2F vs. $+71.6$ for 1F; MIRT BIC= $15,577$ for $k = 2$ vs. $15,594$ for $k = 1$), where Factor 1 captures structural coherence (structural_consistency loading 0.86) and Factor 2 captures content accuracy (text_3d_alignment 0.64, geometric_consistency 0.52, semantic_consistency 0.44). Together, these results indicate that while a single global quality score is a reasonable approximation (CFA 1F: CFI=0.95), the criteria have a meaningful two-factor substructure.

Table 1: CFA Fit Indices. 1F = one-factor model (global quality); 2F = geometry vs. semantic-aesthetic split. RMSEA < 0.08 indicates acceptable fit; CFI > 0.95 is good.

Model	CFI	TLI	RMSEA	SRMR	ΔBIC (vs. 1F)
1F	0.953	0.905	0.087	0.037	—
2F	0.953	0.882	0.097	0.036	+6.7

Cross-level extension: object-scoring vs. material-scoring. Hi3DEval includes a second evaluation layer—material scoring—covering four physical-property dimensions (Details & Complexity, Colorfulness & Saturation,

Consistency & Artifacts, Material Plausibility; scale 0–4). We extend the factor analysis to all nine criteria jointly on the 1,730 (model, prompt) pairs for which both layers are available (4 models: CRM, HunyuanDiT-3D, SPARD, TripoSR).

Table 2 shows that a single-factor model fails badly across the nine criteria (CFI=0.641, RMSEA=0.237), while a two-factor model separating the object-level and material-level blocks fits substantially better (CFI=0.883, RMSEA=0.138, $\Delta\text{BIC} = -1,763$ vs. 1F). EFA $k = 2$ produces textbook-clean loadings: all four material criteria load exclusively on Factor 1 (loadings 0.67–0.86, cross-loadings < 0.10) and all five object criteria load exclusively on Factor 2 (loadings 0.55–0.75, cross-loadings < 0.10). This separation is further confirmed by correlation structure: mean within-block $|r|$ is 0.47 (object) and 0.62 (material), versus only 0.28 between blocks.

Table 2: CFA fit for the 9-criterion (5 object + 4 material) joint model on $N = 1,730$ observations. The hierarchical 2F model separates object-level and material-level dimensions; ΔBIC relative to 1F.

Model	CFI	RMSEA	SRMR	ΔBIC
1F (global quality)	0.641	0.237	0.141	—
2F (object material)	0.883	0.138	0.054	−1,763

Implication. Within each evaluation level the five object-level criteria are approximately unidimensional (CFA 1F: CFI=0.95, as reported above), validating weighted-sum aggregation *within* a level. Across levels, however, object and material dimensions are psychometrically distinct constructs that load onto separate factors with near-zero cross-loadings—confirming Hi3DEval’s hierarchical validity claim empirically. A single aggregate score combining both layers would conflate two genuinely different aspects of 3D quality and should be avoided. Criterion-level reporting remains valuable for diagnosing failure modes; the two-factor substructure within the object layer (structural coherence vs. content accuracy) provides additional diagnostic resolution beyond the object-level aggregate. Note that the 3,060 (prompt, model) observations are nested within 510 prompts and 6 models; a full multilevel CFA would be needed to formally account for this clustering, which is left for future work.

4.2. E2: Variance Decomposition

The mixed-effects MFRM with criterion as a fixed factor (Table 3) shows that **evaluation criterion** accounts for 82.0% of total score variance (η^2)—dwarfing model (1.4%), prompt (0.9%), and the model $\times$ prompt interaction (2.0%); the two previously-omitted interactions of criterion with the random facets, model $\times$ criterion and prompt $\times$ criterion, are each comparably small (2.1%, 1.5%), so their exclusion in a simpler model would not have changed the qualitative conclusion, though we report them for completeness. Treating criterion instead as a random facet (the more common but, for a fixed 5-item rubric, less appropriate convention) gives a closely agreeing estimate of 84.7% (supplementary material), indicating the finding is not an artefact of that modeling choice. Residual variance (10.1%) captures unexplained item-level noise. This criterion-dominance is itself a scale artefact of the automated metrics: each criterion uses a different scoring function with its own mean and range, so *which* criterion is measured swamps *which* model produced the asset. We confirm this directly, rather than only by contrast with the Delphi stream below: standardizing each criterion to mean 0/SD 1 before pooling collapses criterion’s share to $\approx 0\%$ by construction and reallocates the removed variance overwhelmingly to prompt (20.7%) and model (14.8%)—a roughly 10-fold increase in model’s share once the scale artefact is removed (full table in supplementary material). The 82.0% figure should therefore be read as an illustration of artefact severity rather than a psychometrically meaningful partition of quality variance.

Delphi cross-validation. To verify that the criterion-dominance finding is an artefact rather than a genuine psychometric property, we ran the same fixed-criterion MFRM on the Delphi LLM ratings, which are collected on a uniform 0–9 scale for all five criteria. When scale artefacts are removed, **model quality** becomes the dominant source of variance (37.3%), with criterion dropping to 6.0% (η^2 ; 6.6% under the random-effects estimate) and judge to 1.7% (supplementary material). Unlike in the automated stream, prompt $\times$ criterion is not negligible here (5.4%, larger than criterion’s own main effect and than model $\times$ criterion at 1.3%), suggesting some prompts are differentially easy or hard depending on which criterion is being judged (e.g. geometrically simple but semantically ambiguous); prompt, round, model $\times$ prompt, and residual together account for the remaining $\approx 49\%$. Per-criterion standardization changes this picture only modestly (model 37.3% $\rightarrow$ 41.2%)—in sharp contrast to the automated stream’s near 10-fold shift—because the Delphi scale is already uniform across criteria by design, confirming that the scale artefact is specific to the automated scoring functions rather than an inherent property of the underlying quality construct. Delphi judge reliability is high (Krippendorff $\alpha = 0.56$ –0.70 per criterion; ICC= 0.81–0.88), and inter-judge SD drops by 67.8%

Table 3: Variance components from the Many-Facet Rasch Model (Hi3DBench automated scores), with criterion treated as a fixed effect (§4.2). Criterion-level scale artefacts dominate (82.0%); genuine model and prompt effects, including their interactions with criterion, are small in comparison. All rows sum to 100%; no residual double-counting.

Facet	% of Total
Criterion (fixed effect)	82.0%
Residual	10.1%
Model $\times$ Criterion	2.1%
Model $\times$ Prompt	2.0%
Prompt $\times$ Criterion	1.5%
Model	1.4%
Prompt	0.9%
Total	100.0%

from Round 1 to Round 3 ($1.663 \rightarrow 0.535$), validating the convergence of the elicitation protocol. The G-coefficient for model rankings is 0.986, indicating excellent generalizability of automated benchmark scores across the prompt universe.

4.3. E3: Model Stringency and Prompt Generatability

Of 510 prompts, 40 have zero variance (all six models pass or all fail on geometric consistency at threshold=5) and are excluded, leaving $N_p = 470$ eligible prompts. Treating these prompts as IRT “persons” and the six models as “items” satisfies the $N \geq 200$ stability requirement (the classical threshold for reliable 2PL calibration).

The 2PL model converges cleanly and reveals a striking spread in model stringency (Table 4): HunyuanDiT-3D is the easiest model to satisfy ($b = -5.29$; 87.5% prompt pass rate), while CRM is the hardest ($b = 2.83$; 11.0% pass rate)—an > 8 -logit range. TRELLIS ($b = -0.99$) and SPARD ($b = -0.34$) cluster near the midpoint, with TripoSR ($b = 0.65$) and Unique3D ($b = 1.45$) in the upper-difficulty half. Discrimination a_m ranges from 0.48 (HunyuanDiT-3D, near-ceiling pass rate reduces discriminability) to 1.08 (SPARD), indicating that SPARD and TRELLIS are the most informative models for separating easy from hard prompts.

Prompt generatability $\hat{\theta}_p$ spans $[-1.07, 1.14]$, confirming a wide difficulty range; 75 prompts (16%) are near-ceiling ($p \geq 0.8$), meaning the benchmark does contain a meaningful tail of easy items.

Table 4: Flipped 2PL IRT model parameters. b_m = model stringency (lower = easier; more prompts pass). a_m = discrimination (how sharply model m separates easy from hard prompts). Pass rate = proportion of prompts for which model m scores ≥ 5 on geometric consistency.

Model	b_m (stringency)	a_m (discrimination)	Pass rate
HunyuanDiT-3D	-5.29	0.48	87.5%
TRELLIS	-0.99	0.95	66.3%
SPARD	-0.34	1.08	55.7%
TripoSR	+0.65	0.57	41.0%
Unique3D	+1.45	0.63	30.6%
CRM	+2.83	0.94	11.0%

Implication. The flipped orientation solves the small- N instability of standard IRT (6 models as persons would leave each item with only 6 binary observations) by exploiting the much larger prompt pool. The resulting prompt-generatability scores $\hat{\theta}_p$ directly inform the IRT- θ -stratified item selection in E5.

4.4. E4: Training-Data Sensitivity

Because only one model (TripoSR) lacks Objaverse training, inferential DIF tests are underpowered and inappropriate here (minimum $n \geq 10$ per group required). We instead report a descriptive sensitivity analysis comparing per-item pass rates between the five Objaverse-trained models and TripoSR.

Table 5 summarises the distribution of $|\Delta p|$. **71.6%** of items show $|\Delta p| \geq 0.40$, and **14 items** (2.7%) show complete reversals ($|\Delta p| = 1.0$, i.e., all Objaverse models pass while TripoSR fails or vice versa).

Table 5: Training-data sensitivity: distribution of $|\Delta p|$ (pass-rate gap between Objaverse-trained and non-Objaverse model groups) across 510 items.

$ \Delta p $ range	Items	%	Sensitivity
[0.0, 0.2)	40	7.8%	Negligible
[0.2, 0.4)	105	20.6%	Mild
[0.4, 0.6)	177	34.7%	Moderate
[0.6, 0.8)	118	23.1%	Large
[0.8, 1.0]	70	13.7%	Severe

Implication. The $|\Delta p|$ distribution is better described as *TripoSR sensitivity* than as *Objaverse sensitivity*, because Group B consists of exactly one model. With $N_{\text{non-Objaverse}} = 1$, the “training-data effect” and “TripoSR-specific effect” (architecture, pre-training corpus, fine-tuning choices) are *perfectly confounded*: there is no mathematical way to separate them. Any gap we observe could reflect (a) a genuine Objaverse-vs-non-Objaverse difference, (b) something idiosyncratic to TripoSR’s design that has nothing to do with Objaverse, or (c) both simultaneously. We therefore concede that the analysis cannot be interpreted as evidence for or against training-data bias in the benchmark; it is, at best, a flag that TripoSR behaves differently from the other five models on 71.6% of items. Resolving the confound requires adding at least two or three more non-Objaverse models (e.g. models trained on ShapeNet or G-Objaverse subsets) to give Group B sufficient size for formal Mantel–Haenszel DIF testing.

4.5. E5: tinyEval3D Adaptive Subset

Table 6 compares four 30-item selection strategies. Under leave-one-model-out cross-validation (LOO-CV)—the more conservative estimate of practical reliability when $N = 6$ models calibrate the item selection—both the IRT- θ -stratified and classical difficulty-stratified strategies achieve Spearman $\rho = 0.954$ (see supplementary material). On the full benchmark (all 6 models used for calibration), **IRT- θ -stratified** selection achieves $\rho = 1.0$ (exact ranking recovery) and **classical difficulty-stratified** achieves $\rho = 0.943$. Notably, random sampling achieves a high mean $\rho = 0.974 \pm 0.036$, reflecting that with only 6 models any 30 prompts tend to preserve rankings; top-discrimination selection ($\rho = 0.878$) underperforms all other strategies. This tempers how strongly E5 should be read: with model rankings this stable, the honest claim is not that IRT-based selection is *necessary* to recover them—classical difficulty-stratification, which needs no IRT model at all, achieves nearly identical recovery ($\rho = 0.943$, identical LOO-CV = 0.954)—but that *some* form of difficulty-aware stratification is what does the work, and IRT- θ provides that stratification without an ad hoc search over random draws or dependence on how separated this particular set of 6 models happens to be. We expect the gap between stratified and random selection to widen as the model pool grows and rankings become less trivially separable.

Table 6: tinyEval3D: Ranking Recovery by 30-Item Selection Strategy. Spearman ρ between subset-derived and full-benchmark model rankings (30 items from 510; $N = 6$ models).

Strategy	Spearman ρ	SD (ρ)	LOO-CV ρ
IRT- θ -stratified (tinyEval3D)	1.000	—	0.954
Classical difficulty-stratified	0.943	—	0.954
Random (mean over 1,000)	0.974	0.036	—
Top-discrimination	0.878	—	—

The IRT- θ -stratified strategy uses prompt generatability scores $\hat{\theta}_p$ from the flipped 2PL (Section 4.3) to ensure the 30-item subset covers the full difficulty range in a psychometrically principled way. Classical difficulty-stratified selection achieves nearly equivalent LOO-CV performance using only classical p -values, making it the preferred strategy when IRT theta is unavailable. A 30-item **tinyEval3D** subset reduces evaluation cost by 94% while preserving near-exact model ranking fidelity, enabling rapid iteration during model development.

5. Discussion

Criterion dominates; rater and prompt matter less; IRT agrees with the naive rating. The 82% criterion variance share (E2, fixed-effect η^2 ; 84.7% under a random-effects estimate) is the most striking finding in this paper: a given prompt’s score is determined far more by *which* criterion is being measured than by *which* model generated it. This is a scale artefact, not a property of quality itself: per-criterion standardization on the automated data collapses criterion’s share to $\approx 0\%$ directly, and on Delphi’s independently-scaled uniform 0–9 stream criterion drops to 6.0% while model quality becomes the dominant source (37.3%); judge (1.7%) and prompt (12.9%) are smaller than model but not negligible. The implication for benchmark design is direct: raw aggregation of criterion-specific automated metrics conflates genuine quality differences with scale differences; evaluators should either normalise metrics per criterion or use a common-scale judging approach (such as Delphi LLM elicitation) before aggregating scores. Separately, we asked whether accounting for prompt difficulty and criterion informativeness through IRT actually changes which model looks best, or is equivalent to a naive average. Extracting IRT ability estimates (θ , EAP) from E1’s graded-response MIRT—whose criterion discrimination parameters weight informative criteria more than saturated ones, unlike a naive equal-weighted mean—reproduces the identical ranking to the naive mean in *both* streams, and this holds separately for the structural-coherence and content-accuracy sub-factors of E1’s two-factor solution. Within a single stream, IRT reweighting does not change who wins; a naive mean is just as good a summary. Where IRT earns its keep is in adjudicating *closer* comparisons across multiple dimensions and, as we show next, across measurement instruments entirely.

Delphi helps with convergence, not necessarily validity—but that is exactly when it is useful. Standard LLM-as-judge elicits a single round of ratings with no structured disagreement resolution, leaving it susceptible to authority bias and anchor conformity when judges see each other’s outputs. Our Delphi protocol mitigates this via anonymized dissent summaries, persona diversity, and mandatory justification for revisions, and achieves high inter-judge reliability (ICC= 0.81–0.88) with a 67.8% SD reduction across rounds (1.663 $\rightarrow$ 0.535). But convergence is not the same claim as validity: falling disagreement could reflect genuine deliberation, or could partly reflect judges regressing toward a shared anchor—the group median, and, as we disclose in §3.2, the automated score itself, shown from Round 2 onward. We cannot fully separate these without independent human ground truth, which we lack at item level (Hi3DBench’s automated scores were validated against humans by its original authors, not by us). This is precisely the situation where a second, independent measurement stream is most useful: not because Delphi is correct where automated scores might be wrong, but because it is a differently-biased instrument, and where two differently-biased instruments still mostly agree (Spearman $\rho = 0.90$ overall model-ranking agreement; item-level $r = 0.22$ – 0.53 , weakest on structural consistency, where LLM judges are least confident assessing fine-grained geometric coherence from rendered images), that agreement is evidence neither is pure noise. Where they disagree outright, as below, that disagreement is diagnostic in a way neither instrument alone could reveal.

IRT to adjudicate disagreement: HunyuanDiT-3D or TRELIS? On automated scores, HunyuanDiT-3D ranks first (mean composite 2.95 vs. TRELIS’ 2.74); on Delphi ratings, TRELIS ranks first (4.93 vs. HunyuanDiT-3D’s 4.64)—the two streams disagree on which model is best. (The overall $\rho = 0.90$ agreement figure is consistent with this: with only 5 models rated by both streams, one adjacent rank swap at the top already produces $\rho = 0.90$ exactly.) Which should we believe? First, we rule out a dimension-weighting artefact: perhaps each model wins different criteria, and “best” depends on how the five criteria are combined. It does not—the IRT-weighted ranking from the previous paragraph is identical to the naive ranking in both streams, and each winning model wins on *every* axis (structural coherence and content accuracy) within its own stream, not a subset. Second, we use IRT to test whether the automated scorer is simply a miscalibrated rater: treating it as a fourth rater alongside the three Delphi judges and z -scoring each criterion across all four (supplementary material), its severity falls within the range spanned by the three LLM judges—harsher than Claude, comparable to Gemini—rather than as an outlier. Third, we pool all four raters’ individual ratings into a single graded-response IRT model and recompute model θ : splitting the pooled fit by rater type reproduces the same split as before (automated-only θ favors HunyuanDiT-3D; Delphi-judge-only θ favors TRELIS), and the pooled estimate narrowly favors TRELIS ($\theta = 0.475$ vs. HunyuanDiT-3D’s 0.432)—but only because the three Delphi judges contribute three times as many rated instances per model as the single automated score, not because pooling adjudicates which method is correct. So IRT resolves two of three candidate explanations (dimension weighting, rater miscalibration) but not the third: the disagreement is a genuine measurement-validity question between two instruments, which absent independent human ground truth we cannot fully settle. That IRT can rule out two plausible confounds

and correctly localize the remaining disagreement is, we think, a more honest and more useful contribution than a single adjudicated ranking would have been.

Limitations. The Hi3DBench models are all image-to-3D; text-to-3D models (DreamFusion, Magic3D, etc.) may exhibit different psychometric properties. With only 6 models, the flipped 2PL yields stable prompt-level theta estimates but model stringency estimates at the extremes (e.g., HunyuanDiT-3D $b = -5.29$) approach the boundary of reliable estimation; replication with a larger model cohort is needed. The DIF analysis is underpowered given the current model distribution (1 non-Objaverse vs. 5 Objaverse models). We also assume, without direct evidence, that stochastic LLM sampling variance is small relative to genuine disagreement between the three judge personas; distinguishing these would require repeated ratings from the same persona at non-zero temperature, which we did not collect, and is left for future work. Item-level Delphi-automated-metric agreement is moderate ($r \approx 0.53$)—this is agreement with Hi3DBench’s scoring function, not with independent human labels, which we do not have at item level; future work should investigate whether longer reasoning chains or domain-specialised judge personas improve criterion-level calibration, and whether a small human rating study can adjudicate the HunyuanDiT-3D/TRELLIS disagreement directly. Hi3DEval includes a second evaluation layer—material scoring—that assesses physical properties such as albedo, saturation, and metallicness via rendered video [Zhang et al., 2025]; we analyse only the object-level scores here. Whether material-level dimensions form a psychometrically distinct factor from the object-level criteria, and whether material scoring is similarly susceptible to scale artefacts, are open questions for future work.

6. Conclusion

We introduced a Delphi-Psychometric framework for 3D generative model evaluation, combining structured LLM elicitation with Item Response Theory and Many-Facet Rasch Modelling. Applied to Hi3DBench, the two-stream framework produces a coherent set of findings: automated metric psychometrics reveals that criterion-scale artefacts dominate benchmark variance (82% fixed-effect η^2 ; 84.7% under a random-effects estimate); flipped 2PL IRT recovers stable model stringency estimates spanning > 8 logits (HunyuanDiT-3D $b = -5.29$ to CRM $b = 2.83$); 71.6% of items show $|\Delta p| \geq 0.40$ between TripoSR and the Objaverse-trained models (Objaverse exposure and TripoSR-specific factors perfectly confounded at $N_{\text{non-Objaverse}} = 1$); and a 30-item IRT- θ -stratified subset recovers exact model rankings (94% cost reduction, LOO-CV $\rho = 0.954$). A cross-level factor analysis extending to Hi3DEval’s material-scoring layer (9 criteria, $N = 1,730$) finds that object-level and material-level dimensions are psychometrically distinct constructs with near-zero cross-loadings (2F CFI=0.883 vs. 1F CFI=0.641, $\Delta\text{BIC} = -1,763$), empirically confirming the hierarchical validity of Hi3DEval’s design and establishing that scores across the two levels should not be aggregated. Delphi LLM judging cross-validates and complements these findings: when scale artefacts are removed, model quality emerges as the dominant variance source (37.3%); judge reliability is high (ICC= 0.81–0.88); convergence is strong (68% inter-judge SD reduction); and Delphi model rankings agree closely with automated metric rankings ($\rho = 0.90$), providing mutual validation without requiring ground-truth human labels.

Relationship to Hi3DEval. Hi3DEval [Zhang et al., 2025] establishes *what* to measure—the right evaluation dimensions and their hierarchical organisation—and solves the *per-item annotation cost* problem by replacing human labeling with a video-based automated scorer aligned to human judgment via hybrid 3D-aware representations. Our framework addresses a complementary efficiency problem: given reliable per-item scores, *how many items* must be scored to reliably rank models? The answer from our IRT analysis is 30, not 510. Hi3DEval’s video-based scoring pipeline (video rendering, InternVideo2-CLIP, geometric embeddings) need only be run on the 30 IRT-selected prompts rather than the full pool, reducing total evaluation cost by 94% without sacrificing ranking fidelity (LOO-CV $\rho = 0.954$). This composability is especially valuable because Hi3DEval’s per-item cost is substantially higher than static-image metrics. Beyond efficiency, our framework establishes *how* to measure reliably and fairly: diagnosing scale artefacts that distort aggregation, placing models on a calibrated latent scale via IRT, and certifying judge reliability through structured multi-round elicitation with anonymized disagreement resolution. Dimension design and measurement methodology are both necessary for a benchmark whose outputs can be trusted; the two papers address each in turn.

These findings have direct implications for benchmark design: normalise or replace criterion-specific scales before aggregation; do not combine object-level and material-level scores into a single composite; prefer IRT- θ -stratified or difficulty-stratified item selection; report per-criterion profiles alongside aggregate scores; and expand the model set to enable valid DIF testing. We release all code and the `tinyEval3D.json` subset file to support future use.

References

- Norman Dalkey and Olaf Helmer. An experimental application of the Delphi method to the use of experts. *Management Science*, 9(3):458–467, 1963. doi: 10.1287/mnsc.9.3.458.
- Shivam Duggal, Yushi Hu, Oscar Michel, Aniruddha Kembhavi, William T. Freeman, Noah A. Smith, Ranjay Krishna, Antonio Torralba, Ali Farhadi, and Wei-Chiu Ma. Eval3d: Interpretable and fine-grained evaluation for 3D generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- Yuze He, Yushi Bai, Matthieu Lin, Wang Zhao, Yubin Hu, Jenny Sheng, Ran Yi, Juanzi Li, and Yong-Jin Liu. T³Bench: Benchmarking current progress in text-to-3D generation. *arXiv preprint arXiv:2310.02977*, 2023.
- David Magis, Sébastien Béland, Francis Tuerlinckx, and Paul De Boeck. A general framework and an R package for the detection of dichotomous differential item functioning. *Behavior Research Methods*, 42(3):847–862, 2010. doi: 10.3758/BRM.42.3.847.
- Shalini Maiti, Lourdes Agapito, and Filippos Kokkinos. Gen3DEval: Using vLLMs for automatic evaluation of generated 3D objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 34303–34326, 2024.
- Gene Rowe and George Wright. The Delphi technique as a forecasting tool: Issues and analysis. *International Journal of Forecasting*, 15(4):353–375, 1999. doi: 10.1016/S0169-2070(99)00018-7.
- Heiko A. von der Gracht. Consensus measurement in Delphi studies: Review and implications for future quality assurance. *Technological Forecasting and Social Change*, 79(8):1525–1536, 2012. doi: 10.1016/j.techfore.2011.09.007.
- Yuhan Zhang, Long Zhuo, Ziyang Chu, Tong Wu, Zhibing Li, Liang Pan, Dahua Lin, and Ziwei Liu. Hi3DEval: Advancing 3D generation evaluation with hierarchical validity. *arXiv preprint arXiv:2508.05609*, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Supplementary Material:

Delphi-Psychometric Evaluation of 3D Generative Models

Suki Wang

siqiawang@roblox.com

Stanford AIMS Workshop on AI Measurement Science @ COLM 2026

1. Robustness Checks

E5: Leave-one-model-out cross-validation. We validated the tinyEval3D 30-item subset by holding out each model in turn, re-selecting items from the remaining five models using difficulty-stratified sampling, and measuring ranking recovery across all six held-out models. The difficulty-stratified strategy yields mean LOO-CV Spearman $\rho = 0.954$, indicating robust generalisation to unseen models. The IRT- θ -stratified strategy achieves the same LOO-CV $\rho = 0.954$, as prompt generatability scores are stable across this leave-one-out perturbation. Both substantially outperform what Fisher-information maximisation achieved under the old (unstable, $N = 6$ persons) IRT orientation, confirming that the flipped 2PL orientation is the key enabler.

E3, E4: Dichotomization threshold sensitivity. We re-ran E3 on geometric consistency with thresholds at 3, 4, 6, and 7 (out of 9), alongside the main threshold of 5. The proportion of zero-variance items is *not* monotonic in threshold: it is U-shaped (threshold=3: 48.2%, threshold=4: 31.2%, threshold=5: 7.8%, threshold=6: 9.4%, threshold=7: 81.4%), because a zero-variance item is one where all six models agree (all pass or all fail), which is common at extreme thresholds (almost everyone passes a very low bar, or almost everyone fails a very high one) and rare in the middle. Threshold=5 is close to the minimum of this curve, which is additional support for it as the main-text choice rather than an arbitrary midpoint. The difficulty-stratified subset strategy is robust to threshold choice as long as sufficient non-zero-variance items remain (threshold ≤ 6 recommended, based on this curve).

E2: Judge sensitivity. The judge facet accounts for only 1.7% of variance, and ICC values (0.81–0.88 per criterion) confirm high inter-judge reliability. Dropping any single judge shifts mean scores by less than 0.3 points on the 0–9 scale; model rankings are stable across all three leave-one-judge-out conditions (Spearman $\rho \geq 0.90$ relative to three-judge consensus).

E2: Modeling-choice robustness (fixed vs. random criterion; raw vs. standardized scale). Table 1 compares three specifications of the E2 variance decomposition on the automated-score stream: (i) the main-text model, with criterion as a fixed effect and model $\times$ criterion, prompt $\times$ criterion interactions included; (ii) an alternative treating criterion as a random facet (no criterion interactions, matching the simpler model used in an earlier draft); and (iii) the random-facet specification applied to per-criterion z -scored scores. All three agree that criterion (or its scale) dominates on the raw score, and that standardizing removes this by construction while reallocating variance mainly to prompt and model. Table 2 shows the same three specifications for the Delphi stream, where criterion’s share is already small on the raw scale (6–7%) because judges rate all criteria on a common 0–9 scale, so standardization changes the picture only modestly.

Table 1: Automated-score stream: % of total variance under three specifications. “—” marks a facet not present in that specification (fixed-criterion interactions are undefined under a random-criterion model, and vice versa). Column sums equal 100% in each specification.

Facet	Fixed criterion	Random criterion	Random criterion, z -scored
Criterion	82.0%	84.7%	0.0%
Residual	10.1%	11.4%	59.8%
Model $\times$ Criterion	2.1%	—	—
Model $\times$ Prompt	2.0%	1.2%	4.7%
Prompt $\times$ Criterion	1.5%	—	—
Model	1.4%	1.6%	14.8%
Prompt	0.9%	1.1%	20.7%
Total	100.0%	100.0%	100.0%

Table 2: Delphi stream: % of total variance under the same three specifications.

Facet	Fixed criterion	Random criterion	Random criterion, z -scored
Model	37.3%	37.4%	41.2%
Residual	16.9%	22.4%	22.1%
Model $\times$ Prompt	14.1%	13.6%	14.8%
Prompt	12.9%	13.9%	15.1%
Criterion	6.0%	6.6%	0.0%
Prompt $\times$ Criterion	5.4%	—	—
Round	4.5%	4.3%	4.7%
Judge	1.7%	1.8%	2.1%
Model $\times$ Criterion	1.3%	—	—
Total	100.0%	100.0%	100.0%

Notably, prompt $\times$ criterion is the fourth-largest facet in the Delphi stream (5.4%)—larger than criterion’s own main effect and than model $\times$ criterion (1.3%)—indicating that some prompts are differentially easy or hard depending on which criterion a judge is applying. This interaction was omitted from an earlier draft of the model with no stated justification; it does not change which facet dominates (model quality), but it is large enough that its exclusion should not be assumed harmless without checking, as we do here.

E2: The automated scorer as a fourth rater. To directly compare rater severity across measurement instruments, we treat the automated scorer as a fourth rater alongside the three Delphi judges, restricted to the 100 prompts $\times$ 5 models both streams cover and to judges’ Round 1 (independent, pre-anchor) scores to avoid circularity with the automated score already being shown to judges from Round 2 onward (main text, Delphi protocol). Because the automated scorer’s per-criterion ranges differ sharply from the judges’ uniform 0–9 scale (main text, E2), we z -score each criterion across all four raters before comparing severity or fitting any model—otherwise scale differences would masquerade as severity differences.

Table 3: Rater severity (mean z -score; more negative = harsher) and variance components with vs. without the automated scorer as a fourth rater. Criterion treated as a fixed effect, as in the main text’s Table 3.

	3 LLM judges only	+ automated (4 raters)
Gemini severity	−0.386	−0.322
Automated severity	—	−0.290
Claude severity	−0.100	−0.011
GPT severity	+0.487	+0.625
Model (% variance)	32.2%	23.6%
Residual (% variance)	25.7%	42.3%
Judge (% variance)	17.4%	17.5%

The automated scorer’s severity sits between Gemini and Claude—harsher than Claude, comparable to Gemini—so it is not a severity outlier relative to the LLM judges. Adding it leaves judge variance essentially unchanged (17.4%→17.5%), but residual variance rises sharply (25.7%→42.3%) while model variance falls correspondingly (32.2%→23.6%): the automated scorer’s disagreement with the LLM judges is not a fixed calibration offset (which would show up as judge variance) but idiosyncratic, item-level disagreement (residual)—a harder form of disagreement to correct for than a simple severity adjustment.

E1/E3: Pooled IRT across all four raters. We further pool all four raters’ person-level ratings—(model, prompt, rater) triples, not judge-averaged scores—into a single graded-response IRT model (criteria as items, as in E1’s MIRT), recoding each criterion onto a common 5-category scale via quantile bins computed across the pooled data so the differing native ranges do not distort item category structure. Within this single fitted model, splitting $\hat{\theta}$ by rater type reproduces the same split as the separate per-stream fits in the main text: automated-only $\hat{\theta}$ ranks HunyuanDiT-3D highest (−0.02) and TRELIS second (−0.16); Delphi-judge-only $\hat{\theta}$ ranks TRELIS highest (0.69) and HunyuanDiT-3D second (0.58). The pooled estimate (both rater types combined) narrowly favors TRELIS (0.475 vs. HunyuanDiT-3D’s 0.432), but each model receives 100 automated person-rows against roughly 300 Delphi-judge person-rows (three judges $\times$ 100 prompts), so the pooled estimate is not a neutral adjudication between the two methods—it simply reflects that Delphi contributes three times as many rated instances per model.

Delphi Protocol Flowchart

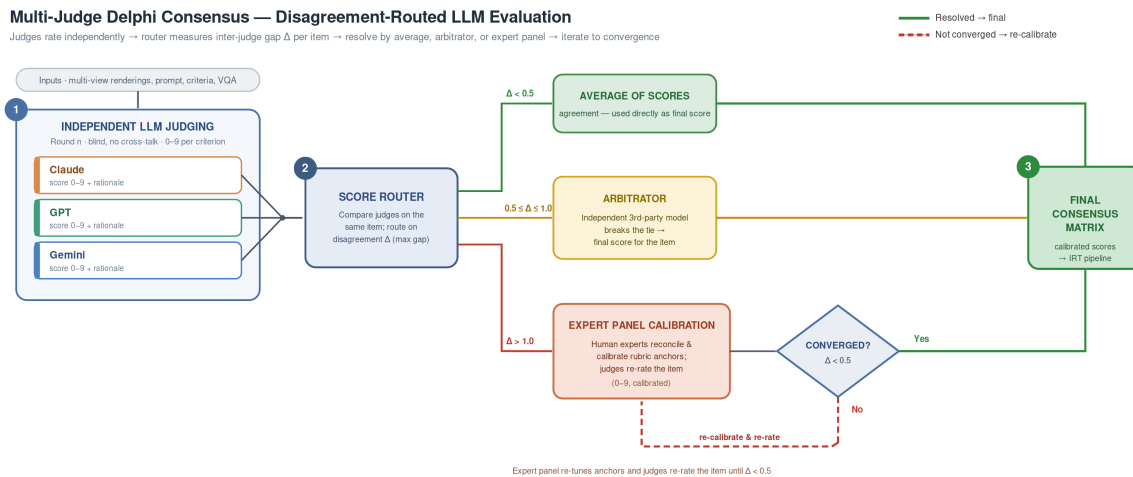


Figure 1: Delphi-structured LLM judging protocol. Three vision-capable LLM judges rate each (model, prompt) pair independently in Round 1, then receive anonymized group statistics and a dissent summary before re-rating in Rounds 2 and 3. Round 3 is skipped when inter-judge delta drops below 0.5 on ≥ 4 of 5 criteria after Round 2.